

III. CONFIGURATION SUMMARY

វិទ្យាស្ថានបច្ចេកវិទ្យាឌីជីថល
INSTITUTE OF DIGITAL TECHNOLOGY

DEVICE SETUP AND IP ADDRESSING

2 Routers

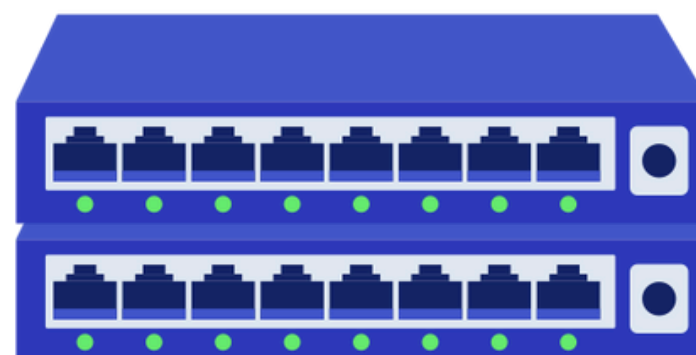
R1 HQ LAN (192.168.1.0/24)

R2 Branch LAN (192.168.2.0/24)



2 Switches

For LAN



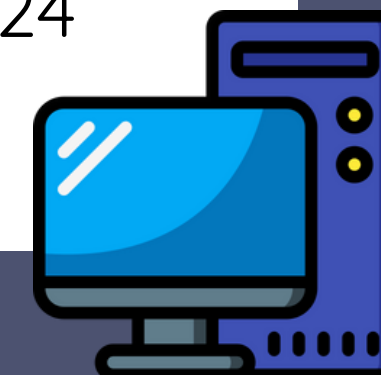
4 PCs

PC1 : 192.168.1.10/24

PC2 : 192.168.1.11/24

PC3 : 192.168.2.10/24

PC4 : 192.168.2.11/24



Main Link (PPP)

- On Serial1/0, uses encapsulation ppp
- R1 as DCE with clock rate 64000
- Handles normal HQ-Branch traffic



Backup Link (ISDN Simulation)

- On Serial1/1, uses encapsulation ppp + DDR
- Key cmds:
 - dialer in-band
 - dialer string
 - dialer-group 1
 - dialer-list 1 protocol ip permit
- backup interface s1/1 auto-activates when PPP fails



MAIN WAN AND BACKUP WAN

ROUTING

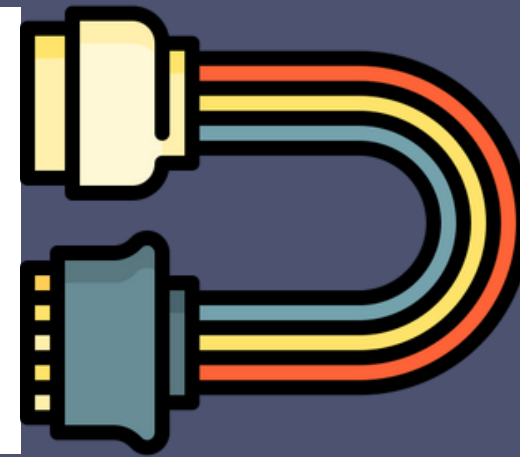
Reach the Branch LAN (192.168.2.0/24) through R2

```
ip route 192.168.2.0 255.255.255.0 10.1.1.2 ← main PPP route  
ip route 192.168.2.0 255.255.255.0 10.1.2.2 200 ← backup ISDN route
```

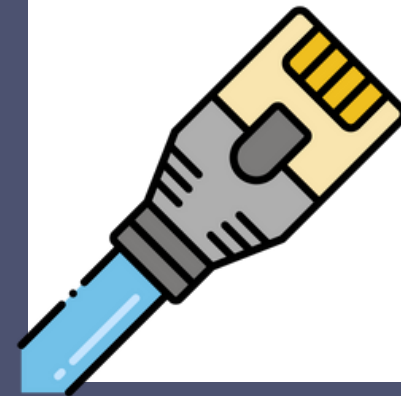
Reach the HQ LAN (192.168.1.0/24) through R1

```
ip route 192.168.1.0 255.255.255.0 10.1.1.1 ← main PPP route  
ip route 192.168.1.0 255.255.255.0 10.1.2.1 200 ← backup ISDN route
```

When PPP link is up, traffic goes through
Serial1/0

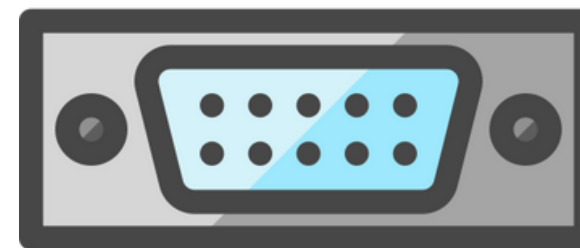


USAGES

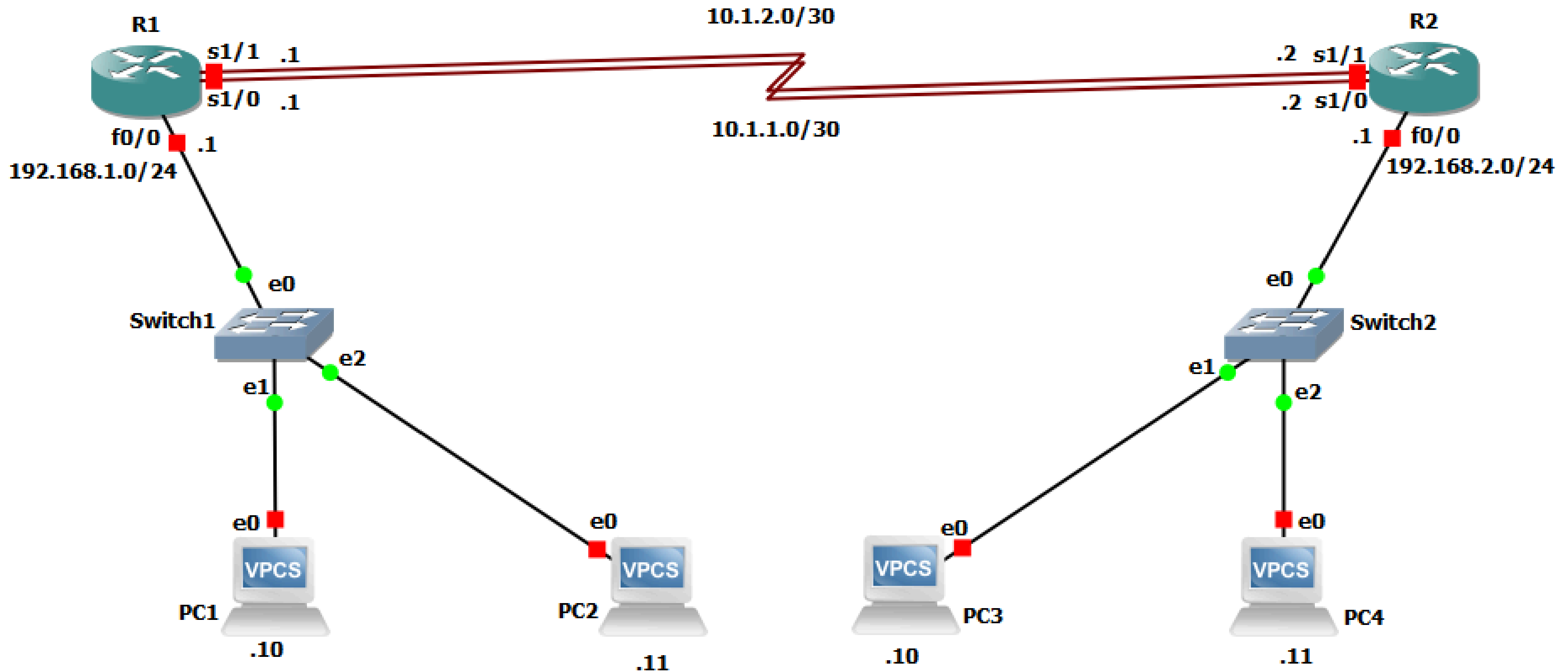


When PPP link fails, ISDN backup link
(Serial1/1) activates automatically

Once PPP comes back, traffic will go
back to normal default



Network Topology



IV. DEMO EXPLANATION

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DEMO VIDEO



[Watch video on YouTube](#)

Error 153

Video player configuration error

